

- a) Differentiate between supervised and unsupervised learning methods using suitable examples.
- b) Explain Decision tree algorithm with example.

- a) Explain the following:
 - i. Supervised learning
 - ii. Unsupervised learning
 - iii. Reinforcement learning
 - iv. Evolutionary learning.
- b) Define Clustering. Illustrate the general Characteristics of Categorization of major Clustering Methods.
- c) Differentiate between K-means and K-medoids Clustering methods with an example.

- a) Describe the various accuracy metrics associated with testing of machine learning algorithms.
- b) Explain Matthew's Correlation Coefficient.
- c) Illustrate Receiver Operating Characteristic (ROC) curve.

- a) Explain normal distribution with a suitable equation.
- b) Describe the Naïve Bayes classifier with appropriate notations.
- c) Illustrate Variance and Covariance with a suitable example.

- a) Describe the following problems with respect to Tasks, Performance and Experience:
 - i. A Checkers learning problem.
 - ii. A Handwritten recognition learning problem.
 - iii. A Weather Prediction problem.
- b) Explain the steps in designing a learning system in detail.
- c) Explain different types of machine learning with an example.

- b) Explain Matthew's Correlation Coefficient.
- c) Consider the Dataset given below:

Rid	age	income	student	Class: buys_computer
1	youth	high	no	no
2	youth	high	no	no
3	middle_aged	high	no	yes
4	senior	medium	no	yes
5	senior	low	yes	yes
6	senior	low	yes	no
7	senior	low	yes	yes

Given a new Data tuple with **age = "senior "**, **income = "low "** and **student = "yes"**, Classify the Data tuple using Naïve Bayesian Classifiers.

- a) Explain:
 - i. Confusion Matrix.
 - ii. Recall.
 - iii. Accuracy.
 - iv. Sensitivity.
- b) Explain Variance and Covariance with example.
- c) Illustrate Bias-Variance Trade-off.

- a) Name any two Supervised Learning Algorithms. With examples justify why they belong to the category of 'Supervised' Learning?
- b) In a Medical Classification Model to diagnose if a patient is diabetic or not 50,000 patients were examined. The number of TP = 8000, FP = 5000, TN = 35,000, FN = 2000. Do the following for this Medical Classification Model:
- Draw the Confusion Matrix
 - Calculate the Accuracy value
 - Assess the Sensitivity Value
 - Estimate the Specificity Value
 - Measure the Error Rate
 - Determine the Precision Value
 - Compute the Recall Value
 - Evaluate the F-Measure
 - Is the data sample given to you a Balanced or an Imbalanced one? Justify
 - Is this Model a 'High Precision' or 'High Recall' model? Justify
- c) Explain with an example the difference between Overfitting and Underfitting of the ML Model.
- a) What do you mean by Imbalanced Datasets? With relevant examples explain why Accuracy fails as a performance metrics for these datasets?
- b) Explain Correlation Coefficient. How 'correlated' are the parameters 'Temperature' and 'Ice-Cream Sales' for the following information given:

Temperature	Ice Cream Sales
66	8
72	11
77	15
84	20
83	21
71	11
65	8
75	10

- c) With an appropriate example explain Bayesian Rule. Derive Bayesian Rule from Conditional Probabilities.

- a) Illustrate the multilayer perceptron algorithm along with an example.
- b) What is a perceptron? Discuss the perceptron for the XOR function along with a table.
- c) Discuss the advantages and disadvantages of the following Gradient Descent Algorithm:
- Mini Batch Gradient Descent
 - Stochastic Gradient Descent.
- a) China's task is to diagnose 100 patients in Wuhan with Corona Virus present in 50% of its general population. Chinese data analysts assume a black box model, where they put in information about patients and receive a score between 0 and 1. They can alter the threshold for labeling a patient as positive (has the disease) to maximize the classifier performance. They will evaluate thresholds from 0.0 to 1.0 in increments of 0.1, at each step calculating the **Precision, Recall** and **F1 Score**. Following are the classification outcomes at each threshold.

Threshold	TP	FP	TN	FN
0.0	50	50	0	0
0.1	48	47	3	2
0.2	47	40	9	4
0.3	45	31	16	8
0.4	44	23	22	11
0.5	42	16	29	13
0.6	36	12	34	18
0.7	30	11	38	21
0.8	20	4	43	33
0.9	12	3	45	40
1.0	0	0	50	50

Which is the Threshold at which the overall F1 Score is the highest?

- b) Explain with an example each the difference between the following:
- Overfitting and Underfitting of the ML Model
- a) Apply Naive Bayes Classifier to the below dataset and determine if your Height is Tall, Hair is Red and Eyes are Brown, would you be attractive?

Height	Hair	Eyes	Attractive?
Small	Blonde	Brown	No
Tall	Dark	Brown	No
Tall	Blonde	Blue	Yes
Tall	Dark	Blue	No
Small	Dark	Blue	No
Tall	Red	Blue	Yes
Tall	Blonde	Brown	No
Small	Blonde	Blue	Yes

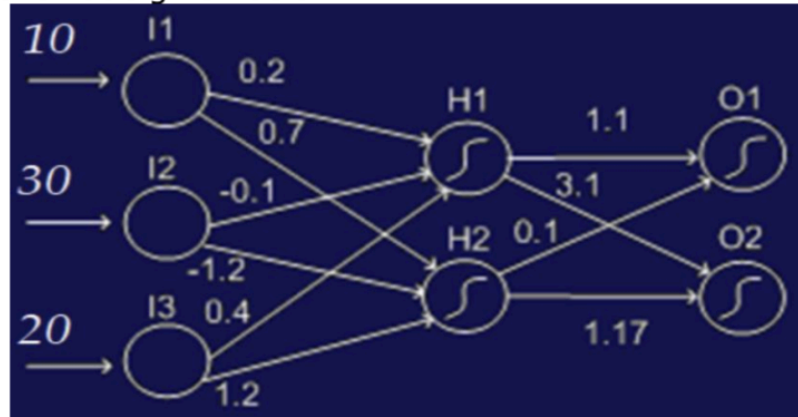
- b) What is the difference between:
- (i) Supervised and Unsupervised ML Algorithms
 - (ii) Classification & Regression ML Algorithms.
- c) What do you mean by Imbalanced Datasets? Why does Accuracy fail as a performance metrics for these datasets?
- a) Illustrate and explain your understanding of Gradient Descent? With examples explain the different types of Gradient Descent.
- b) Study the MLP given below:

MLP Schematic	The MLP Configuration Values are as follows			
	w13=3	w14=6	w23=4	w25=5
	w35=2	w45=4	Bias Weight at Hidden Layer U4=-6	Bias Weight at Output Layer U5=-3.93
	Input X1=1	Input X2=0	Bias Weight at Hidden Layer U3=1	Target Ytarget = 1 Bias Value U0=1

Calculate the following:

- (i) Total Error(E_{total}) after Forward Pass at Output Layer
- (ii) Updated weight of w35
- (iii) Updated weight of w13

- a) Calculate the Total Error(E_{total}) after **Forward Pass** at Output Layer for the MLP Schematic given below.



- b) Design a two-input perceptron that implements **either of** the boolean functions
- (i) A **AND** B
 - (ii) A **OR** B
- b) Consider the following set of points: $\{(-2, -1), (1, 1), (3, 2)\}$
- i) Find the **least square regression line** for the given data points.
 - ii) Plot the given points
 - iii) Plot the regression line in the same rectangular system of axes as the points
 - iv) Predict the value for a new data point, 2.
- c) Compare and Contrast:
- (i) Simple and Multiple Linear Regression
 - (ii) Multi-Class and Multi-Label Classification
- a) Give **at least four** points of difference between **Linear** and **Logistic** Regression.

- a) In the below case studies, determine which measure among '*High Precision*' or '*High Recall*' would you use for each of them. Justify your answer.

- A) People are innocent until proven guilty as per Indian Law. We want to avoid false convictions, even at the cost of criminals running free.
- B) In Planet Utopia, the law demands that the number of criminals running free is brought as minimum as possible, even at the cost of wrongly convicting innocent people.

For *either* of the case studies given above, assume the values of TP, TN, FP, FN and calculate the following:

(i) Confusion Matrix (ii) Accuracy (iii) Error Rate (iv) Sensitivity (v) Specificity (vi) Precision (vii) Recall (viii) F-Measure.

- a) Write and describe the Perceptron Algorithm. List its assumptions and limitations.
- a) What is Linear Regression? List the critical assumptions of Linear Regression. Mention few applications of Linear Regression.
- b) Design a Perceptron that produces output of logical OR of their inputs.
- c) What do you understand by 'Perceptron Activation Function'? What is its purpose? What are the types?
- d) With examples and diagrams explain what do you understand by the following:
- i. ROC
 - ii. Perceptron Convergence and Linear Separability
 - iii. MAP.
- b) Answer the following:
- (i) Difference between Train, Validation and Test Dataset
 - (ii) What is the relation between Imbalanced Dataset and Accuracy as a performance measure?
 - (iii) What is the relation between Overfitting, Underfitting of ML models with the terms 'Bias' and 'Variance'.

- b) Describe the Learning Phase and Testing Phase for Naive Bayes Classifier algorithm. Apply Naive Bayes Classifier to the below dataset and classify if a customer will buy a product on a holiday, with discount and free delivery.

	A	B	C	D
1	Day	Discount	Free Delivery	Purchase
2	Weekday	Yes	Yes	Yes
3	Weekday	Yes	Yes	Yes
4	Weekday	No	No	No
5	Holiday	Yes	Yes	Yes
6	Weekend	Yes	Yes	Yes
7	Holiday	No	No	No
8	Weekend	Yes	No	Yes
9	Weekday	Yes	Yes	Yes
10	Weekend	Yes	Yes	Yes
11	Holiday	Yes	Yes	Yes
12	Holiday	No	Yes	Yes
13	Holiday	No	No	No
14	Weekend	Yes	Yes	Yes
15	Holiday	Yes	Yes	Yes

Naive_Bayes_Dataset